

Can XAI Methods Reliably Identify What a Model Actually Learned, or Do They Reflect Dataset Bias and Architecture Artefacts?

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Abstract

Explainability in Artificial Intelligence is crucial for understanding and interacting with deep-learning-based models. Yet many popular XAI methods are significantly affected by a model's internal biases and architectural limitations, raising a fundamental question: do these explanations actually reveal what a model learned, or do they merely reflect what the explanation algorithm was designed to find? To investigate this, we present a systematic evaluation of four XAI methods—Gradient-weighted Class Activation Mapping (GradCAM), Integrated Gradients (IntGrad), LIME, and KernelSHAP—applied to four neural architectures: VGG-16, ResNet-50, ViT-B16, and BagNet-33. Across 320 attribution maps on the Imagenette benchmark, we measure five quantitative metrics (faithfulness, AOPC, sufficiency, sparsity, stability), run cascading parameter randomization sanity checks, and analyse both cross-architecture attribution agreement and metric independence. The central finding is that XAI method choice drives explanation quality $5.8\times$ more than architecture choice (ANOVA $F = 21.50$ by method vs. $F = 3.70$ by architecture). Used together, IntGrad and LIME form the most reliable combination for explaining feature contributions—though both are influenced by the model's architecture. BagNet-33, a locally-constrained architecture with a 33×33 -pixel receptive field, achieves the highest faithfulness in the study (IntGrad: 0.701) and near-pixel GradCAM precision (sparsity: 0.048), while partially correcting the spurious correlations that mislead all three conventional architectures. ALL resources supporting this work are available on https://github.com/sankalp567/DL_project

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Neural networks*; *Machine learning*.

*All authors contributed equally to this research.

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Keywords

Explainable AI, GradCAM, Integrated Gradients, LIME, KernelSHAP, Faithfulness, AOPC, Attribution, Dataset Bias, BagNet, Vision Transformer

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1 Introduction

The gap between a model's prediction accuracy and a human's ability to trust that prediction has motivated a decade of work in Explainable AI [2]. As neural networks now achieve human-level performance on many classification tasks, the question of *why* a model makes a particular decision has become as important as whether the decision is correct. In safety-critical settings—medical imaging, autonomous vehicles, legal decision support—a plausible-looking explanation that does not reflect the model's actual reasoning process may be more dangerous than having no explanation at all, because it creates unwarranted confidence in a wrong model.

Many XAI methods have been proposed to address this need, yet their reliability is far from guaranteed [3]. Explanations can be shaped by biases in the training data, by architectural choices made when building the model, or by the internal assumptions of the explanation algorithm itself. Without rigorous evaluation, it is difficult to know which of these factors is driving an explanation—or whether the explanation reflects anything meaningful at all.

This paper addresses that uncertainty with a structured comparative study, asking: *Can XAI methods reliably identify what a model actually learned, or do they reflect dataset bias and architecture artefacts?* We apply GradCAM, Integrated Gradients, LIME, and KernelSHAP to four architectures spanning CNNs and Vision Transformers, and evaluate all combinations on five independent quantitative metrics. Beyond accuracy, we test whether explanations survive sanity checks, whether they agree across architectures, and whether our metric set itself captures independent evaluation dimensions.

Our main contributions are as follows:

- A five-metric evaluation framework with ANOVA, applied uniformly across four architectures and four XAI methods on Imagenette;

- Empirical evidence that method choice drives explanation quality $5.8\times$ more than architecture ($F = 21.50$ vs. $F = 3.70$);
- Demonstration that IntGrad and LIME together can expose dataset-level spurious correlations that model confidence scores conceal;
- The first evaluation of GradCAM on BagNet-33, showing that a constrained receptive field restores GradCAM precision and partially corrects dataset bias;
- Validation that 9 of 10 metric pairs are genuinely independent ($|r| < 0.30$), confirming the framework measures non-redundant aspects of explanation quality.

2 Related Work

2.1 Gradient-Based Attribution

GradCAM [15] generates class-discriminative spatial maps by weighting convolutional activations by their globally averaged gradient with respect to the class score. Its maps are coarse by design, bounded by the receptive field of the chosen convolutional layer, and require no modifications to the model architecture. Integrated Gradients [18] takes a different approach, accumulating gradients along a linear path from a reference (baseline) input to the actual input. The resulting pixel-level attributions satisfy the completeness axiom—one of the few principled guarantees in the XAI literature—and tend to be far more spatially focused than GradCAM.

2.2 Perturbation-Based Attribution

Rather than using gradients, LIME [13] fits a local linear surrogate model to predictions on masked versions of the input, requiring only black-box access to the model. KernelSHAP [12] extends this idea to game-theoretic Shapley values, computed efficiently via weighted linear regression. Both methods operate at superpixel granularity, which limits their spatial resolution but makes them applicable to any model.

2.3 XAI Evaluation

The most influential critique of XAI evaluation came from [3], who showed that several widely-used saliency methods produce nearly identical maps regardless of whether the model weights are randomised—a result that calls into question whether these explanations reflect model learning at all. [14] introduced the Area Over the Perturbation Curve (AOPC) as a quantitative faithfulness measure, while [19] formalised the complementary notions of faithfulness (necessity) and sufficiency. Explanation stability under input perturbations was studied by [4], who found high variance in stochastic methods.

2.4 Architectures and Interpretability

Two architectural developments are particularly relevant here. [5] introduced BagNet, whose class logits are computed as a sum of independent patch-level contributions, making the model interpretable by construction. [8] introduced Vision Transformers, which process images as token sequences with global self-attention—a representation fundamentally different from the spatial feature maps that GradCAM was designed to read. Although GradCAM's limitations on transformers have been noted [15], they have not

previously been quantified with a multi-metric evaluation framework.

3 Methodology

3.1 Dataset and Preprocessing

All experiments use Imagenette [10], a 10-class subset of ImageNet covering tench, English springer, cassette player, chain saw, church, French horn, garbage truck, gas pump, golf ball, and parachute. Images are resized to 224×224 and normalised with standard ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). We select 20 validation images via stratified sampling (2 per class), ensuring all 10 classes are represented in every comparison.

3.2 Model Architectures

VGG-16 [16]. A sequential 16-layer CNN with no skip connections, pretrained on ImageNet-1K (top-1: 71.6%). Its large receptive fields at the final convolutional layers tend to produce spatially diffuse attribution maps.

ResNet-50 [9]. A 50-layer residual network, pretrained on ImageNet-1K (top-1: 76.1%). Skip connections enable a deeper feature hierarchy and more structured spatial activations than VGG-16.

ViT-B16 [8]. A Vision Transformer that splits images into 16×16 patches encoded as 197 tokens (196 patch tokens plus one CLS token). The CLS token aggregates global context for classification. Pretrained on ImageNet-1K (top-1: 81.1%).

BagNet-33 [5]. A modified ResNet whose effective receptive field is constrained to 33×33 pixels per patch. Class logits are the sum of independent patch-level contributions, $f(x) = \sum_i q(x_i)$, making the model interpretable by construction. Pretrained on ImageNet-1K.

3.3 XAI Methods

GradCAM. For VGG-16 and BagNet-33 we use the final convolutional layer; for ResNet-50, layer4. For ViT-B16 we implement a manual hook-based variant that strips the CLS token from the final encoder output, reshapes the 196 remaining patch tokens to a 14×14 grid, and bilinearly upsamples to full resolution. All maps are min-max normalised to [0, 1].

Integrated Gradients. Computed via Captum [11] with a zero baseline and $n_{\text{steps}} = 75$ integration steps. Attribution is the absolute value of the channel mean, then normalised. The zero baseline ensures the completeness axiom holds.

LIME. LimeImageExplainer with $n_{\text{samples}} = 750$ and SLIC superpixel segmentation. Predictions are computed in batches of 64 to avoid GPU memory errors. The signed 20-superpixel mask is used directly as the attribution map.

KernelSHAP. shap.KernelExplainer with approximately 50 SLIC superpixels as binary features, an all-grey reference image, and $n_{\text{samples}} = 750$ Shapley approximation samples. Inference is chunked into batches of 32.

3.4 Evaluation Metrics

Faithfulness. We remove the top- $k\%$ most-attributed pixels and measure the resulting drop in model confidence, averaged over

$k \in \{10, 20, 30\}$:

$$\text{Faith} = \frac{1}{3} \sum_{k \in \{0.1, 0.2, 0.3\}} [f_c(x) - f_c(x \odot m_k)] \quad (1)$$

where $f_c(x)$ is the softmax probability for the predicted class and m_k retains the bottom- $(1-k)\%$ of pixels.

AOPC [14]. A 10-step deletion curve spanning 5% to 50% of pixels, with attribution indices sorted once and reused for computational efficiency:

$$\text{AOPC} = \frac{1}{10} \sum_{k=1}^{10} [f_c(x) - f_c(x \odot m_{0.05k})] \quad (2)$$

Averaging over ten deletion fractions makes AOPC more statistically stable than three-point faithfulness.

Sufficiency. Rather than removing the most attributed pixels, we keep only them and measure how much confidence is retained:

$$\text{Suff} = \frac{1}{3} \sum_{k \in \{0.1, 0.2, 0.3\}} f_c(x \odot \tilde{m}_k) \quad (3)$$

Together with faithfulness, this distinguishes pixels that are *necessary* from those that are also *sufficient* for the prediction.

Sparsity. The fraction of pixels whose normalised attribution exceeds threshold $\tau = 0.5$. Lower sparsity means more focused and interpretable localisation; higher sparsity means the attribution is spread across most of the image.

Stability. Mean cosine similarity between the original attribution map and maps produced from 10 Gaussian-noise-perturbed inputs ($\sigma = 0.05$):

$$\text{Stab} = \frac{1}{n} \sum_{i=1}^n \frac{a(x) \cdot a(x + \epsilon_i)}{\|a(x)\| \|a(x + \epsilon_i)\| + \epsilon} \quad (4)$$

A reliable explanation should not change substantially under small perturbations.

3.5 Sanity Checks

Following [3], we progressively randomise model weights layer by layer (sampling every 4th parameter layer to span the full depth), recording the Pearson correlation between the original attribution map and the map produced by the increasingly randomised model at each step. A trustworthy method should show $r \rightarrow 0$ as the model loses its learned representations.

3.6 Attribution Agreement and Statistical Testing

To test whether methods capture image semantics or model-specific patterns, we compute pairwise Spearman r between attribution maps from the same XAI method applied to different architectures on the same image. Separately, one-way ANOVA tests whether metric means differ significantly across XAI methods and across architectures. Global Pearson r between all $\binom{5}{2} = 10$ metric pairs determines whether the evaluation framework is internally redundant.

4 Results

4.1 Faithfulness and AOPC

Figure 1 and Table 1 summarise faithfulness across all 16 architecture–method combinations. IntGrad consistently produces the highest faithfulness on every architecture; the gap is particularly stark on ViT-B16, where IntGrad (0.428) is more than four times higher than GradCAM (0.093). BagNet-33 IntGrad (0.701) is the best individual result in the study—15% above the best CNN score (VGG-16 IntGrad: 0.610). At the other extreme, KernelSHAP on ResNet-50 averages only 0.048, and in 15 of 80 KernelSHAP records (18.75%), removing the most-attributed pixels actually *increases* model confidence—directly falsifying the explanation. AOPC values follow the same ranking throughout.

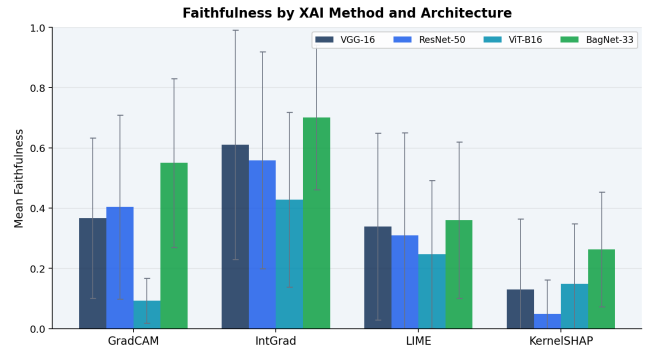


Figure 1: Mean faithfulness by XAI method and architecture. Error bars show standard deviation across 20 images. IntGrad leads on every architecture; BagNet-33 IntGrad achieves the study’s highest score (0.701).

Table 1: Mean Faithfulness (std) by Architecture and XAI Method

Arch.	GradCAM	IntGrad	LIME	KernelSHAP
VGG-16	0.367 (0.266)	0.610 (0.380)	0.338 (0.310)	0.130 (0.235)
ResNet-50	0.404 (0.305)	0.559 (0.360)	0.309 (0.341)	0.048 (0.113)
ViT-B16	0.093 (0.075)	0.428 (0.290)	0.247 (0.245)	0.148 (0.200)
BagNet-33	0.550	0.701	0.360	0.263

Bold: highest per row. BagNet std not available.

4.2 Necessity vs Sufficiency: The Attribution Gap

A striking pattern emerges when faithfulness and sufficiency are placed side by side (Figure 2, Table 2): IntGrad’s maps are highly faithful but nearly useless when retained in isolation, while GradCAM’s coarser maps retain enough context to partially reconstruct the prediction on their own.

The interpretation is straightforward: IntGrad identifies the small set of pixels whose removal most damages the model, but those pixels alone are not enough to reconstruct the prediction. Decisions

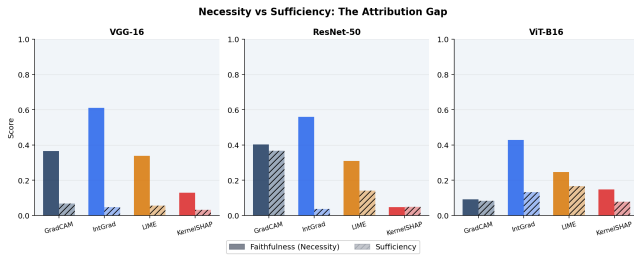


Figure 2: Faithfulness (necessity) vs sufficiency for the three main architectures. IntGrad finds pixels that are necessary but insufficient; GradCAM’s broader coverage preserves enough spatial context that masking everything else still yields partial confidence.

Table 2: Mean Sufficiency by Architecture and XAI Method

Arch.	GradCAM	IntGrad	LIME	KernelSHAP
VGG-16	0.068	0.047	0.056	0.033
ResNet-50	0.367	0.037	0.142	0.050
ViT-B16	0.084	0.133	0.167	0.079

are distributed across the image in a way that no single attribution map fully captures. GradCAM’s broad, spatially diffuse maps inadvertently cover enough of the image that, when everything else is zeroed out, the model can still maintain some confidence from the retained region.

4.3 Sparsity

VGG-16 GradCAM (sparsity 0.772) highlights 77% of pixels—essentially covering the entire image and offering no meaningful spatial guidance. At the other end of the spectrum, BagNet-33 GradCAM (0.048) produces maps almost as focused as IntGrad. As discussed in Section 6, this is a direct consequence of BagNet’s constrained receptive field, not an accidental property of this specific model or test set.

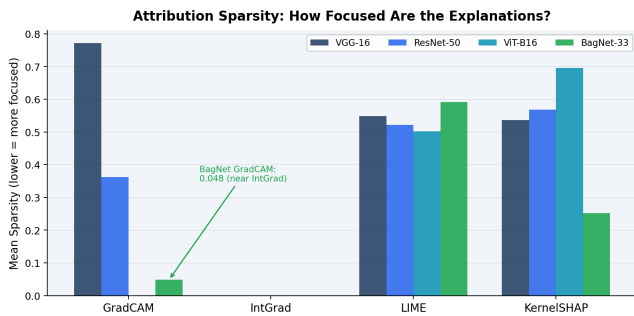


Figure 3: Attribution sparsity by method and architecture. Lower is more focused. BagNet-33 GradCAM (0.048) approaches IntGrad-level precision thanks to its localised receptive field.

Table 3: Mean Sparsity by Architecture and XAI Method (lower = more focused)

Arch.	GradCAM	IntGrad	LIME	KernelSHAP
VGG-16	0.772	0.001	0.548	0.537
ResNet-50	0.362	0.001	0.522	0.568
ViT-B16	0.000*	0.001	0.502	0.695
BagNet-33	0.048	0.001	0.591	0.252

*All-zero maps: sparsity=0 is not true precision.

4.4 Stability

Figure 4 and Table 4 show stability results. GradCAM is near-perfectly stable on all CNN architectures (VGG-16: 0.996, ResNet-50: 0.997, BagNet-33: 0.983), which follows directly from the deterministic nature of gradient computation—add a small amount of noise and the gradient map barely changes. KernelSHAP is the least stable across all architectures (0.156–0.725), consistent with its stochastic superpixel sampling. The ViT-B16 GradCAM entry of 0.000 is not a measure of instability; it reflects undefined cosine similarity between zero vectors, because the attribution maps are uniformly empty.

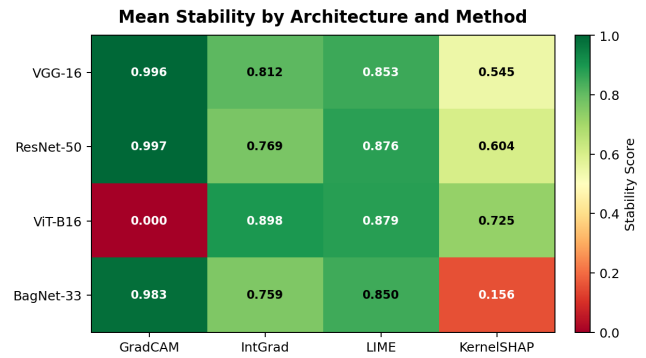


Figure 4: Stability across architectures and methods. Green = high stability; red = low. GradCAM is deterministically stable on CNNs. ViT-B16 GradCAM (0.000) is undefined rather than genuinely unstable.

Table 4: Mean Stability by Architecture and XAI Method

Arch.	GradCAM	IntGrad	LIME	KernelSHAP
VGG-16	0.996	0.812	0.853	0.545
ResNet-50	0.997	0.769	0.876	0.604
ViT-B16	0.000*	0.898	0.879	0.725
BagNet-33	0.983	0.759	0.850	0.156

*Undefined cosine similarity (zero maps).

4.5 ANOVA: Method vs Architecture Effects

Table 5 and Figure 5 present the headline finding of this work. All five metrics vary significantly by XAI method (all $p < 0.01$). By architecture, four of five metrics also show significance. The one exception—AOPC by architecture ($F = 2.78$, $p = 0.064$)—is notable: AOPC is the most rigorous deletion metric, and it does not produce a significant architecture effect. In other words, *what method you choose matters far more than what model you evaluate it on*. Switching from KernelSHAP to IntGrad improves faithfulness by up to 10×, regardless of which architecture is used.

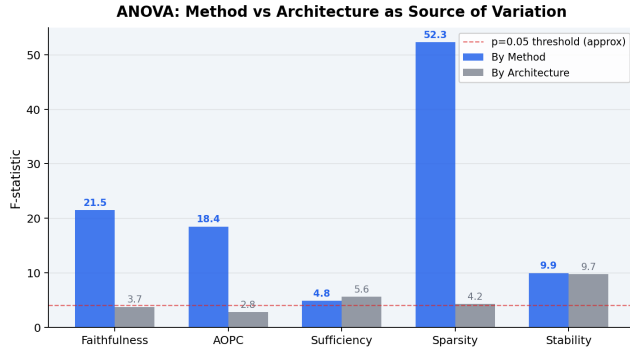


Figure 5: ANOVA F -statistics by method (blue) and architecture (grey). Method effects consistently dominate. The dashed red line marks the approximate $p = 0.05$ threshold.

Table 5: One-Way ANOVA: F -Statistics and Significance

Factor	Metric	F	p	Sig.
By Method	Faithfulness	21.50	<0.001	***
By Method	AOPC	18.45	<0.001	***
By Method	Sufficiency	4.85	0.003	**
By Method	Sparsity	52.32	<0.001	***
By Method	Stability	9.92	<0.001	***
By Architecture	Faithfulness	3.70	0.026	*
By Architecture	AOPC	2.78	0.064	—
By Architecture	Sufficiency	5.56	0.004	**
By Architecture	Sparsity	4.25	0.015	*
By Architecture	Stability	9.70	<0.001	***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, —: not significant.

4.6 Metric Independence

Before interpreting any results in depth, it is worth asking whether the five metrics are actually measuring different things. Figure 6 answers that question. The only redundant pair is Faithfulness–AOPC ($r = 0.967$)—both measure deletion-based quality, just at different granularities. All nine remaining pairs have $|r| < 0.30$, confirming that each of the remaining metrics captures a genuinely distinct aspect of explanation quality.

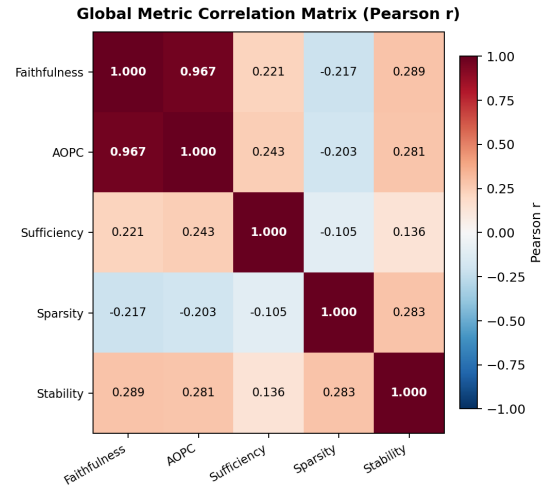


Figure 6: Global Pearson r between evaluation metrics, pooled across all architectures, methods, and images. Faithfulness↔AOPC is the only redundant pair ($r = 0.967$). All others are independent ($|r| < 0.30$), validating the five-metric framework.

4.7 Attribution Agreement

Figure 7 shows whether methods produce similar attributions regardless of which model they are applied to. IntGrad achieves moderate cross-architecture agreement on all three pairs (Spearman r : 0.420–0.497), suggesting its attributions partly reflect the image content rather than just the model’s internal structure. GradCAM–ViT pairs are undefined due to zero maps; KernelSHAP shows near-zero agreement (0.030–0.196), consistent with its instability.

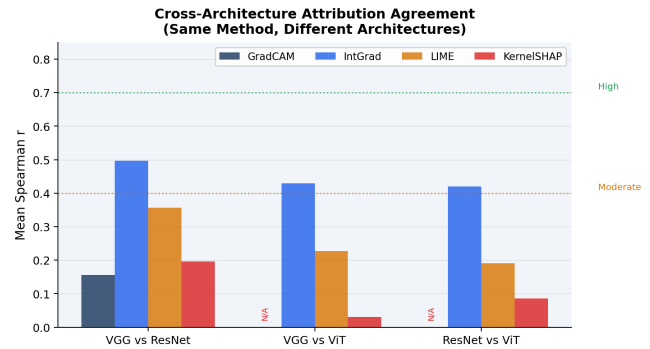


Figure 7: Cross-architecture Spearman r for the same method on different architectures. IntGrad shows moderate input-driven agreement. KernelSHAP shows near-zero agreement. GradCAM–ViT pairs are undefined (N/A).

4.8 Faithfulness Ranking and Sanity Checks

Figure 8 ranks all valid architecture–method combinations by faithfulness. BagNet-33 occupies four of the top seven positions, confirming that its interpretable-by-design structure improves multiple XAI methods simultaneously.

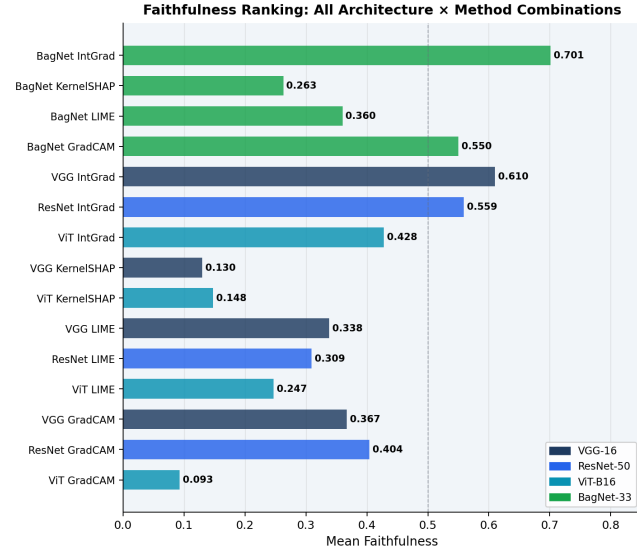


Figure 8: Faithfulness ranking across all architecture×method combinations. BagNet-33 (green) occupies four of the top seven positions. ViT-B16 GradCAM is the lowest valid score (0.093).

The sanity check results (Figure 9, Table 6) reveal architecture-specific failure modes that raw metrics alone would miss. ViT-B16 GradCAM produces NaN Pearson r at every randomisation step, because the attribution maps are zero from the start. ViT-B16 IntGrad shows $r = 0.724$ at the CLS token layer, meaning that randomising the mechanism that actually drives classification barely changes the explanation. BagNet-33 IntGrad shows flat $r \approx 0.42$ at all depths—expected given that all layers contribute equally to the patch-additive logit.

Table 6: Cascading Parameter Randomisation: Summary

Arch.	Method	r Pattern	Interpretation
ResNet-50	GradCAM	0.717→0.186	conv1 insensitive
ResNet-50	IntGrad	0.282→0.303	Moderate decay
VGG-16	GradCAM	−0.73→0.10	Good sensitivity
VGG-16	IntGrad	−0.01→0.20	Good sensitivity
ViT-B16	GradCAM	All NaN	Complete failure
ViT-B16	IntGrad	0.724→0.163	CLS token insensitive
BagNet-33	IntGrad	Flat ≈ 0.42	No hierarchical decay
All	LIME	Decays to low	Good (model-agnostic)

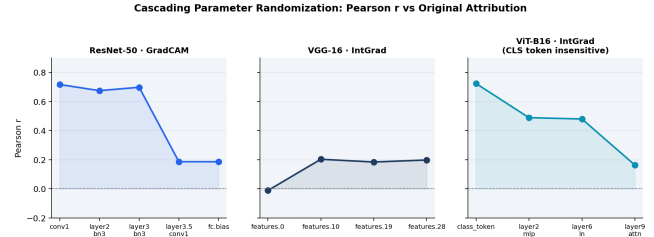
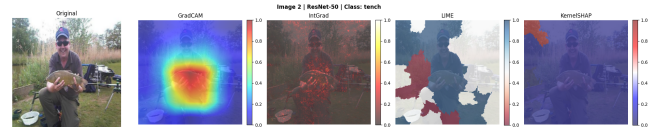


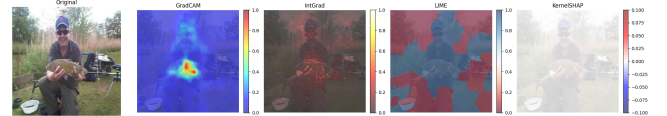
Figure 9: Cascading parameter randomisation for three representative cases. Left: ResNet-50 GradCAM stays high at early layers (conv1: 0.717) because GradCAM reads from layer4 and early-layer changes propagate weakly. Centre: VGG-16 IntGrad drops quickly, indicating genuine model sensitivity. Right: ViT-B16 IntGrad stays high at the CLS token layer (0.724), revealing the gradient path bypasses the primary classification mechanism.

5 Qualitative Case Studies

5.1 Case Study 1: Tench — Bias Hidden in Plain Sight



(a) ResNet-50: both GradCAM and IntGrad focus on the angler's body, not the fish. Confidence: 1.000. Faithfulness: 0.954 (GradCAM), 0.979 (IntGrad).



(b) BagNet-33: GradCAM shifts to the fish body. Faithfulness: 0.952. The 33-pixel receptive field cannot encode the full-scene co-occurrence that misleads the CNN.

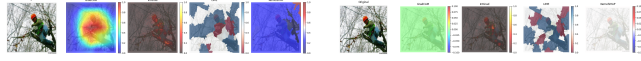
Figure 10: Tench (Image 2): Spurious correlation exposed by IntGrad and LIME in CNNs, and partially corrected by BagNet-33.

Figure 10 is the most consequential result in the study. Both ResNet-50 and VGG-16 predict tench with perfect confidence (1.000), and both GradCAM and IntGrad point to the angler holding the fish—not to the fish itself. LIME reaches the same conclusion independently. All three methods are technically faithful: removing the angler really does drop model confidence. The problem is not the explanation; it is the model. The CNN learned to associate “person with fishing gear in an outdoor setting” with tench, rather than learning any visual property of the fish.

BagNet-33 tells a different story. Its GradCAM correctly localises the fish body (faithfulness 0.952). With only a 33-pixel window available, the model cannot simultaneously encode the full scene,

so it is forced to learn local features of the fish itself. This is the clearest evidence in our study that architecture design can directly influence what a model learns, and that XAI methods are sensitive enough to detect the difference.

5.2 Case Study 2: Chain Saw — When Architecture Matters for Bias



(a) ResNet-50: focus on the worker (faith: 0.161). (b) ViT-B16 IntGrad shifts to the blade (faith: 0.862).

Figure 11: Chain saw (Image 7): ResNet-50 has learned that workers in safety equipment indicate chain saws. ViT-B16 attends globally and correctly identifies the tool itself.

Here the attribution lands on a person in safety equipment, not the chain saw he is operating. ResNet-50 GradCAM and IntGrad both attribute to the worker's body (faithfulness 0.161). ViT-B16 IntGrad, however, attributes to the chain saw blade area and achieves faithfulness 0.862—the largest single cross-architecture faithfulness gap in the dataset (0.701). The transformer's global self-attention is not constrained to look locally, so it integrates context from the entire scene and identifies the actual discriminating object. This is a case where ViT's architecture actively corrects a bias that CNN local filters reinforce.

5.3 Case Study 3: Gas Pump — When Text Becomes the Feature

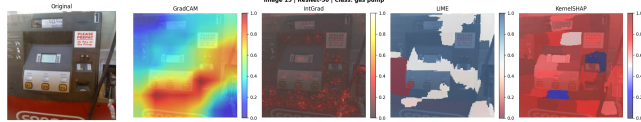


Figure 12: Gas pump (Image 15): ResNet-50 GradCAM and IntGrad concentrate on the “PLEASE PREPAY” sign, not the pump structure. IntGrad faithfulness: 0.884. The model learned to identify gas pumps by their signage.

GradCAM and IntGrad both highlight the payment-related text in Figure 12. IntGrad faithfulness is 0.884—technically correct, but the explanation reveals a troubling shortcut: the model treats “PLEASE PREPAY” as a defining feature of gas pumps, not the pump itself. When the same evaluation is run on an antique hand-crank pump with no signage (Image 16, confidence 0.770), faithfulness collapses to near-zero across all methods. Without the text cue, the model has no fallback representation.

5.4 Case Study 4: GradCAM Failure on ViT-B16

The uniform field in Figure 13 is not noise or a configuration error—it appears on every single one of our 20 test images when GradCAM is applied to ViT-B16. The root cause is a mismatch of assumptions: GradCAM reads convolutional spatial feature maps ($B \times C \times H \times W$),

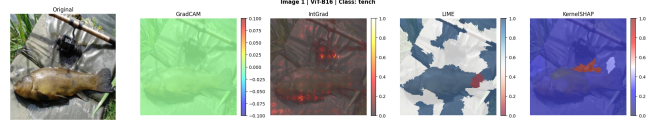


Figure 13: ViT-B16 (Image 1, tench): GradCAM produces a uniform map across all 20 images—sparsity = 0, stability = 0, sanity = NaN. IntGrad still produces a meaningful attribution. The GradCAM failure is architectural, not a software issue.

while ViT's final encoder outputs a token sequence ($1 \times 197 \times 768$). The ReLU in GradCAM's computation eliminates all signal when applied to attention-based representations, because the weighted sums are uniformly negative at the last encoder layer. No hyper-parameter choice can fix this; it is a fundamental incompatibility between method and architecture.

5.5 Case Study 5: Correct Feature Learning

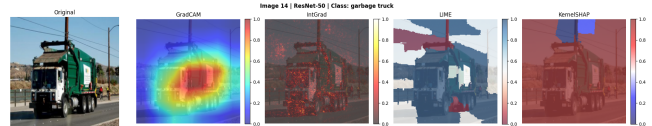


Figure 14: Garbage truck (Image 14): ResNet-50 GradCAM correctly centres on the rear compactor mechanism. All four methods broadly agree. Faithfulness: 0.951. Sufficiency: 0.367 (highest in the study).

Not every result is a failure. Figure 14 shows ResNet-50's attribution on a cleanly photographed garbage truck: GradCAM, IntGrad, LIME, and KernelSHAP all converge on the rear compactor—the mechanically distinctive part of the vehicle. Faithfulness is 0.951 and sufficiency is 0.367, the highest in the study; the compactor region alone nearly sustains the model's confidence. When the image is unambiguous and the model has learned the right feature, multiple XAI methods agree, and the metrics confirm it.

6 Discussion

6.1 Using IntGrad and LIME Together

No single method dominates across all conditions, but IntGrad and LIME together offer the most reliable picture of what a model has learned. IntGrad is sensitive to individual feature contributions—it traces gradients back through the network to find the pixels whose presence or absence most affects the output. LIME takes a broader view, fitting a surrogate model to regions of the image and highlighting those that collectively influence the prediction. When both methods point to the same region, the agreement is meaningful: it shows that the model's decision is robust across two entirely different algorithmic perspectives. When they diverge, the divergence is itself informative—usually indicating a spurious correlate that only one method captures, or an architectural mismatch.

Both methods are bounded by the architecture they are applied to, however. IntGrad on ViT-B16 misses the CLS token aggregation step, as the sanity check confirms ($r = 0.724$ at that layer). LIME's

superpixel boundaries do not align with BagNet's 33-pixel patches. There is no architecture-agnostic ground truth to appeal to; an IntGrad map on ResNet-50 and one on ViT-B16 tell different stories about the same image, and both are faithful to their respective models. Faithfulness is the standard, and meeting it is the best that current XAI can offer.

6.2 Context-Awareness vs Feature Sensitivity Across Architectures

ViT-B16 is more context-aware: its global self-attention integrates information across the entire image before making a decision. CNN models like ResNet-50 and VGG-16 are more locally sensitive, building representations hierarchically from small patches. BagNet-33 is designed to bridge the gap—looking locally (33 pixels at a time) but making its final decision through a sum of patch votes, giving each region an equal and interpretable contribution.

This difference shows up directly in the faithfulness comparison. On images where the discriminating content is spread across the scene—a worker with a chain saw, a church half-hidden in fog—ViT-B16 IntGrad faithfulness significantly exceeds CNN faithfulness. On close-up, single-object images—a French horn filling the frame, a golf ball on a green—CNNs outperform ViT. The faithfulness gap between architectures is essentially a measure of how well the architecture's inductive biases match the image's spatial structure.

6.3 Why GradCAM Precision Depends on Receptive Field

Looking across architectures, GradCAM sparsity follows a clear pattern: VGG-16 (0.772) $\rightarrow$ ResNet-50 (0.362) $\rightarrow$ BagNet-33 (0.048). The more the architecture constrains how much of the image a single activation can "see," the more focused GradCAM becomes. VGG-16's final-layer activations are influenced by the entire image, so GradCAM highlights nearly everything. BagNet-33's hard 33-pixel limit forces gradient concentration to the specific patches responsible for each class logit, yielding maps almost as sharp as IntGrad. GradCAM's well-known coarseness is not an inherent limitation of the method—it is a reflection of the architecture it is reading.

6.4 KernelSHAP: A Reliability Caveat

KernelSHAP ranks last or near-last on faithfulness across three of four architectures, produces the highest sparsity variance, and generates negative faithfulness in 15 of 80 cases (18.75%). These problems trace back to the fundamental difficulty of approximating Shapley values: with 50 superpixels as features, the coalition space has 2^{50} elements, and 750 samples cover a negligible fraction of it. Published work suggests 10,000–50,000 samples for stable SHAP estimates [7]. Until compute budgets allow that level of sampling, LIME is the more reliable perturbation-based alternative.

6.5 Explanations Fail When Models Are Uncertain

Across all low-confidence predictions in the dataset, all four methods produce near-zero or negative faithfulness. This is not specific to any method or architecture—it is a property of post-hoc XAI in

general. When a model has no confident internal representation of a class, there are no high-attribution pixels to find. Explanations degrade exactly when users would most want to scrutinise a prediction.

6.6 Answering the Research Question

Can XAI methods reliably identify what a model actually learned, or do they reflect dataset bias and architecture artefacts?

The honest answer is: **partially, and only under the right conditions**. IntGrad and LIME together can faithfully expose what a model has learned, including spurious correlations that confidence scores would never surface. But being faithful to the model is not the same as being semantically correct. A faithful explanation of a model that learned to recognise tench by the angler holding it is still a reliable explanation of wrong learning. XAI reveals bias; it does not correct it.

Architectural artefacts introduce a different kind of failure: GradCAM on ViT-B16 does not produce an incorrect explanation—it produces no explanation at all. Standard metrics detect this failure automatically (sparsity = 0, stability = 0, sanity = NaN), but only if they are actually computed. Finally, explanations degrade under uncertainty in a way that no method selection can prevent.

The partial exception is BagNet-33. By constraining the receptive field architecturally, it simultaneously improves GradCAM precision, reduces the spurious correlations that mislead both CNNs, and achieves the highest faithfulness in the study. Interpretable-by-design architecture is a more durable path to trustworthy XAI than post-hoc method selection alone.

7 Limitations

Dataset size. Twenty images is sufficient to support comparative conclusions but not per-class statistical tests. Imagenette is also an unusually easy benchmark for pretrained ImageNet models—over 90% of predictions exceed 0.90 confidence—which prevents meaningful confidence-stratified analysis.

Hyperparameters. We use $n_{\text{steps}} = 75$ for IntGrad and $n_{\text{samples}} = 750$ for perturbation-based methods, both below published best-practice thresholds (300 and 10,000, respectively). Results should be treated as comparative rather than absolute.

Single baseline. IntGrad is evaluated with a zero baseline only. Alternative baselines (mean-image, Gaussian blur [17]) may yield substantially different maps, particularly for non-photographic images such as the chain saw product diagram (Image 8).

ViT XAI alternatives. We document GradCAM's failure on ViT-B16 but do not evaluate attention rollout [1] or DINO self-attention [6], which are architecturally appropriate alternatives for transformer-based models.

BagNet accuracy tradeoff. BagNet-33 offers improved interpretability at the cost of lower top-1 accuracy relative to ResNet-50 on full ImageNet. This tradeoff is acknowledged but not quantified here.

8 Conclusion

Across 320 attribution evaluations spanning four architectures and four XAI methods, a consistent picture emerges. The choice of XAI method matters far more than the choice of model architecture (ANOVA $F = 21.50$ vs. $F = 3.70$). IntGrad and LIME together are

the most reliable combination, with IntGrad tracing necessary features and LIME independently validating which regions influence the prediction. Used together, they successfully expose spurious correlations—trench predicted from the angler, chain saw from the worker, gas pump from payment text—that high model confidence would otherwise conceal.

Architectural effects shape explanations in ways that method selection alone cannot overcome. GradCAM is incompatible with ViT-B16 and produces no attribution on any of the 20 test images. BagNet-33, by contrast, constrains the receptive field architecturally and thereby improves both GradCAM precision and XAI reliability across multiple methods simultaneously, achieving the highest faithfulness in the study (IntGrad: 0.701) and near-pixel GradCAM localisation (sparsity: 0.048).

The one limitation that no method or architecture can solve is uncertainty: when the model is not confident, the explanation degrades in parallel. Post-hoc XAI is bounded by the quality of the model it explains.

Taken together, these findings argue for treating XAI method selection as a primary design decision rather than an afterthought, for testing method–architecture compatibility as a prerequisite before deployment, and for considering interpretable-by-design architectures as a first-class option wherever reliability matters.

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